

Session 11 (AIML) -Assignment Question

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- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
session 11.py > ...
1  try:
2      a = float(input("Enter first number: "))
3      b = float(input("Enter second number: "))
4
5      result = a / b
6      print("Division =", result)
7
8  except ValueError:
9      print("Please enter valid numeric values.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\ebugpy-2026.6.0-win32-x64\bundled\libs\de
Enter first number: 10
Enter second number: a
Please enter valid numeric values.
PS C:\Users\tanve\Desktop\ITR> |
```

Q2.

PROGRAM:

```
session 11.py > ...
1  try:
2      num = int(input("Enter an integer: "))
3
4      a = float(input("Enter first number: "))
5      b = float(input("Enter second number: "))
6
7      print("Division =", a / b)
8
9  except ValueError:
10     print("ValueError: Please enter valid numeric values.")
11
12 except ZeroDivisionError:
13     print("ZeroDivisionError: Cannot divide by zero.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd
ebugpy-2026.6.0-win32-x64\bundled\libs
Enter an integer: 25
Enter first number: 2
Enter second number: 5
Division = 0.4
PS C:\Users\tanve\Desktop\ITR> |
```

Q3.

PROGRAM:

```
session 11.py > ...
1  try:
2      num = int(input("Enter an integer: "))
3
4      a = float(input("Enter first number: "))
5      b = float(input("Enter second number: "))
6
7      print("Division =", a / b)
8
9  except ValueError:
10     print("ValueError: Please enter valid numeric values.")
11
12 except ZeroDivisionError:
13     print("ZeroDivisionError: Cannot divide by zero.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::; cd 'c:\ebugpy-2026.6.0-win32-x64\bundled\libs\de
Enter an integer: 55
Enter first number: 25
Enter second number: 3
Division = 8.333333333333334
PS C:\Users\tanve\Desktop\ITR> 
```

Q4.

PROGRAM:

```
session 11.py > ...
1  try:
2      a = float(input("Enter first number: "))
3      b = float(input("Enter second number: "))
4
5      result = a / b
6
7  except ValueError:
8      print("Invalid input.")
9
10 except ZeroDivisionError:
11     print("Cannot divide by zero.")
12
13 else:
14     print("Division =", result)
15     print("Division performed successfully!")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\
ebugpy-2026.6.0-win32-x64\bundled\libs\debu
Enter first number: 12
Enter second number: 2
Division = 6.0
Division performed successfully!
PS C:\Users\tanve\Desktop\ITR>
```

Q5.

PROGRAM:

```
session 11.py > ...
1  try:
2      a = float(input("Enter first number: "))
3      b = float(input("Enter second number: "))
4
5      result = a / b
6      print("Division =", result)
7
8  except ValueError:
9      print("Invalid input.")
10
11 except ZeroDivisionError:
12     print("Cannot divide by zero.")
13
14 finally:
15     # This block always executes
16     print("Program execution completed.")
```

OUTPUT:

```
● PS C:\Users\tanve\Desktop\ITR> & 'C:\Users\tanve\A
s\debugpy\launcher' '53839' '--' 'c:\Users\tanve\De
Enter first number: 25
Enter second number: 3.2
Division = 7.8125
Program execution completed.
○ PS C:\Users\tanve\Desktop\ITR> |
```

Q6.

PROGRAM:

```
session 11.py > ...
2     a = float(input("Enter first number: "))
3     b = float(input("Enter second number: "))
4
5     result = a / b
6
7 except ValueError:
8     print("Invalid input.")
9
10 except ZeroDivisionError:
11     print("Division by zero is not allowed.")
12
13 else:
14     print("Division =", result)
15     print("Operation successful!")
16
17 finally:
18     print("Thank you for using the program.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher'
Enter first number: 13.8
Enter second number: 6.7
Division = 2.0597014925373136
Operation successful!
Thank you for using the program.
PS C:\Users\tanve\Desktop\ITR>
```

Q7.

PROGRAM:

```
session 11.py > ...
1  try:
2      age = int(input("Enter your age: "))
3
4      if age < 0:
5          raise ValueError("Age cannot be negative")
6
7      print("Your age is:", age)
8
9  except ValueError as e:
10     print("Error:", e)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\ebugpy-2026.6.0-win32-x64\bundled\libs\de
Enter your age: 18
Your age is: 18
PS C:\Users\tanve\Desktop\ITR> |
```

Q8.

PROGRAM:

```
session 11.py > ...  
1  try:  
2      num = int(input("Enter a number: "))  
3      result = 100 / num  
4      print("Result =", result)  
5  
6  except (ValueError, ZeroDivisionError):  
7      print("Error: Enter a valid non-zero number.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:;  
ebugpy-2026.6.0-win32-x64\bundled\l  
Enter a number: 87  
Result = 1.1494252873563218  
PS C:\Users\tanve\Desktop\ITR>
```

Q9.

PROGRAM:

```
session 11.py > ...
1  try:
2      num = int(input("Enter a number: "))
3      result = 100 / num
4      print("Result:", result)
5
6  except ValueError:
7      print("Please enter a valid number")
8
9  except ZeroDivisionError:
10     print("Cannot divide by zero")
11
12 except Exception:
13     print("Some error occurred")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c::;
ebugpy-2026.6.0-win32-x64\bundled\I
Enter a number: 89
Result: 1.1235955056179776
PS C:\Users\tanve\Desktop\ITR> |
```

Q10.

PROGRAM:

```
session 11.py > ...
1  while True:
2      print("\n----- Calculator -----")
3      print("1. Addition")
4      print("2. Subtraction")
5      print("3. Multiplication")
6      print("4. Division")
7      print("5. Exit")
8
9      choice = input("Enter your choice (1-5): ")
10
11     if choice == "5":
12         print("Exiting Calculator...")
13         break
14
15     try:
16         num1 = float(input("Enter first number: "))
17         num2 = float(input("Enter second number: "))
18
19         if choice == "1":
20             print("Result =", num1 + num2)
21
22         elif choice == "2":
23             print("Result =", num1 - num2)
24
25         elif choice == "3":
26             print("Result =", num1 * num2)
27
28         elif choice == "4":
29             print("Result =", num1 / num2)
30
31         else:
32             print("Invalid choice!")
33
34     except ValueError:
35         print("Invalid input! Please enter numbers only.")
36
37     except ZeroDivisionError:
38         print("Division by zero is not allowed.")
39
40     finally:
41         print("Operation attempted.")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\ITR> c:; cd 'c:\Users\tanve\Desktop\ITR\debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher'
----- Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 1
Enter first number: 46
Enter second number: 67
Result = 113.0
Operation attempted.

----- Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter first number: 40
Enter second number: 80
Result = -40.0
Operation attempted.

----- Calculator -----
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
```

```
Enter your choice (1-5): 3
Enter first number: 55
Enter second number: 8
Result = 440.0
Operation attempted.
```

```
----- Calculator -----
```

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

```
Enter your choice (1-5): 4
Enter first number: 66
Enter second number: 78
Result = 0.8461538461538461
Operation attempted.
```

```
----- Calculator -----
```

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

```
Enter your choice (1-5): 5
Exiting Calculator...
```